

OPTIMIZATION FUNDAMENTALS

Standard Forms: General: $\min f(x)$ s.t. $g_i(x) \leq b_i, h_j(x) = c_j, x \in X$ **LP Standard:** $\min c^T x$ s.t. $Ax = b, x \geq 0$

Linear Algebra (Past Exam Focus): Matrix rank: Count linearly independent rows/columns **Example:** $A = \begin{bmatrix} 0 & 3 & -2 & -1 \\ 4 & -9 & 1 & 3 \\ -6 & -15 & -1 & 5 \end{bmatrix}$ has rank = 2

System $Ax = b: -m = n$, full rank: unique solution - $m < n$: infinite solutions - $m > n$: over-determined - Inconsistent: $\text{rank}(A) < \text{rank}([A|b])$

Gradients & Hessians (Past Exam): $\nabla f = [\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}]^T$
 $\nabla^2 f = [\frac{\partial^2 f}{\partial x_i \partial x_j}]$

Example 1: $f(x_1, x_2) = \sqrt{x_1 - 1} + (x_2 - 1)^{1/4}$ at $(3, 2)$ $\nabla f = [\frac{1}{2\sqrt{2}}, \frac{1}{4}]$

Example 2: $f(x_1, x_2) = 2(x_1 - 3)^2 + 6x_1x_2 + 2(x_2 + 3)^2 + 6x_1 + 2x_2$ at $(2, 3)$ $H = \begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$

Taylor Expansions: 1st order: $f(x) \approx f(x_0) + \nabla f(x_0)^T(x - x_0)$ **Example:** $f(x, y) = x^3 - 2xy + y^2$ at $(-2, 3)$ gives $13x - 3y - 6$

Optimality Conditions: 1st order necessary: $\nabla f(x^*) = 0$ **2nd order (at stationary x^*):** - $\nabla^2 f(x^*) \succ 0$ (positive definite) $\Rightarrow$ strict local min - $\nabla^2 f(x^*) \succeq 0$ but not PD (positive semidefinite) $\Rightarrow$ inconclusive (flat directions; possible min) - $\nabla^2 f(x^*) \prec 0$ (negative definite) $\Rightarrow$ strict local max - $\nabla^2 f(x^*) \preceq 0$ but not ND (negative semidefinite) $\Rightarrow$ inconclusive (flat directions; possible max) - $\nabla^2 f(x^*)$ indefinite $\Rightarrow$ saddle (neither min nor max)

Matrix definiteness test: For $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$: PD if $a > 0, \det > 0$; PSD if $a \geq 0, \det \geq 0$

CONVEXITY (Major Exam Topic)

Convex Sets: Definition: $\lambda x + (1 - \lambda)y \in S \forall x, y \in S, \lambda \in [0, 1]$

Examples from past exams: - $\{x \in \mathbb{R}^2 | \min\{x_1, x_2\} \geq 1\}$: NOT convex - $\{x \in \mathbb{R}^2 | x_1 \geq \ln x_2, x_2 > 0\}$: NOT convex - $\{x \in \mathbb{R}^2 | x_1^2 + x_2^2 \geq 4\}$: NOT convex - Hyperplane $\{x | a^T x = b\}$: convex - Halfspace $\{x | a^T x \leq b\}$: convex

Set Properties (Past Exam): $X = \{x \in [-1, 1]^5 : \sum_{i=1}^5 x_i \geq -2\}$: - **Bounded:** Yes (box constraints) - **Closed:** Yes (closed inequality)

Convex Functions: Definition: $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$

Tests: 1. **2nd order:** $\nabla^2 f(x) \succeq 0$ (PSD) 2. **1st order:** $f(y) \geq f(x) + \nabla f(x)^T(y - x)$

Examples from exams: - $f(x_1, x_2) = 2|x_1 + 1| + |x_2 - x_1|$: convex - $f(x) = -2x^2 + 6x + 10$: NOT convex (concave) - $f(x) = x^5$ on $[-1, 1]$: NOT convex - $f(x) = \ln(x^2)$ on $(0, \infty)$: convex

CONVEX OPTIMIZATION PROBLEMS

Problem Classification (Past Exam): Convex problem: $\min f(x)$ s.t. $g_i(x) \leq b_i, h_j(x) = c_j$ where f, g_i convex, h_j linear
Examples from past exams: - $\min\{x^4 + y^4 : x^2 + y^2 \geq 20\}$: NOT convex (feasible region not convex) - $\max\{\cos(x) : x \in [0, 2\pi]\}$: NOT convex (cos not concave on interval) - $\min\{\ln(x) : x \geq 1\}$: convex - $\min\{-x - 8y : x + 7y \geq 10, e^x \leq 1\}$: convex

LINEAR PROGRAMMING

Standard Form Conversion: From: $\max\{2x_1 - 3x_2 : x_1 + x_2 \leq 2, x_1 + 2x_2 \geq 3\}$ **To:** $\min\{-2x_1^+ + 2x_1^- + 3x_2^+ - 3x_2^- : x_1^+ - x_1^- + x_2^+ - x_2^- + s_1 = 2, x_1^+ - x_1^- + 2x_2^+ - 2x_2^- - s_2 = 3, \text{all} \geq 0\}$

Basic Solutions & Extreme Points: Basic feasible solution (BFS): Extreme point of feasible region **Degenerate BFS:** Some basic variables = 0 **Optimality:** If LP has optimal solution, then has optimal BFS

Simplex Method: Steps: Start BFS $\rightarrow$ Check optimality $\rightarrow$ Choose entering variable $\rightarrow$ Ratio test $\rightarrow$ Update basis **Reduced costs:** For min problem, optimal if all $\bar{c}_j \geq 0$ **Degeneracy example:** Reduced costs $(3, -1, 0)$ with ratios $(1, 0) \rightarrow$ degenerate

TRANSPORTATION PROBLEM

Standard Form: $\min \sum_{i=1}^m \sum_{j=1}^n c_{ij}x_{ij}$ s.t. $\sum_{j=1}^n x_{ij} \leq s_i$ (supply), $\sum_{i=1}^m x_{ij} \geq d_j$ (demand), $x_{ij} \geq 0$

Infeasibility condition: Total supply $\sum s_i < \max\{d_j\}$

SPECIAL FORMULATIONS

Absolute Value Problems (Past Exam): $\min |x - 2| + 3|1 - x|$ s.t. $x \geq 0 \rightarrow \min z_1 + 3z_2$ s.t. $z_1 \geq x - 2, z_1 \geq -(x - 2), z_2 \geq 1 - x, z_2 \geq -(1 - x), x \geq 0$

Max/Min Functions (Past Exam): $\min\{\frac{1}{2}(|x - 4| + |x + 5|) + \frac{1}{2}\max\{|x - 4|, |x + 5|\}\} \rightarrow \min \frac{1}{2}z_1 + \frac{1}{2}z_2$ s.t. $z_1 \geq |x - 4| + |x + 5|, z_2 \geq |x - 4|, z_2 \geq |x + 5|$

SPECIFIC CALCULATIONS

Feasible Point Tests (Past Exam): $X = \{x \in \mathbb{R}^2 | x_1 - x_2 \leq -1, 2x_1 - x_2 \leq 0, x_2 \geq -\frac{1}{2}\}$ Test $(0, -\frac{1}{2})$: $0 - (-\frac{1}{2}) = \frac{1}{2} \not\leq -1 \rightarrow$ NOT feasible

Optimization Outcomes (Past Exam): $\max\{412x - 511y : 0 \leq x \leq 1, y \leq 1\}$ Since coefficients positive for x , negative for y : optimal at $x = 1, y = 0$

Piecewise Linear Functions: $f(x) = \min\{10x - 1, 2x + 10, 20\}$: Can be reformulated as LP $f(x) = \sum_{i=1}^{2019} 2^{-i} |e^{i/2} x - \sin(i)|$: NOT standard LP form

DECISION CONCEPTS

Timing Decisions: Here-and-now: Decision before uncertainty realized **Wait-and-see:** Decision after uncertainty realized

Function Properties: Level sets: Convex function does NOT always have convex level sets **Boundary optimization:** Max of convex function over compact set achieved at boundary **Uniqueness:** Non-convex problems can have multiple optimal solutions

KEY ABBREVIATIONS LP=Linear Programming, BFS=Basic Feasible Solution, PD=Positive Definite, PSD=Positive Semidefinite